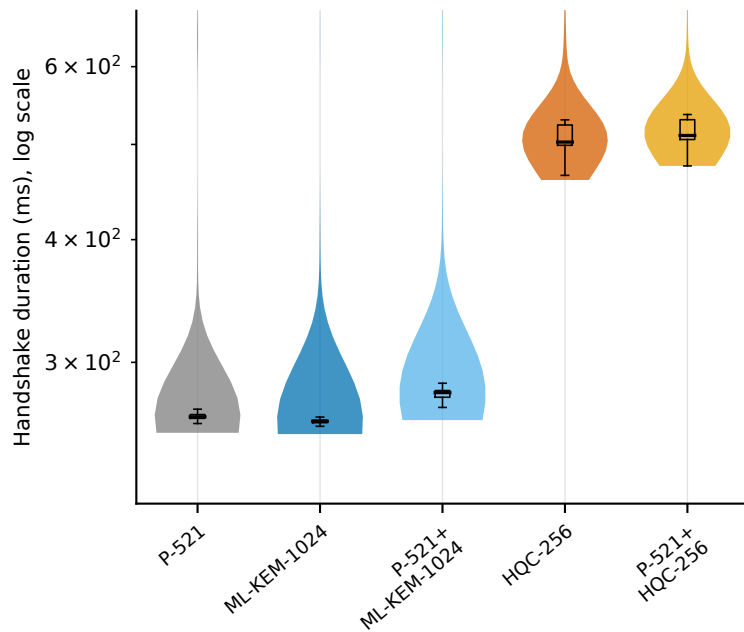
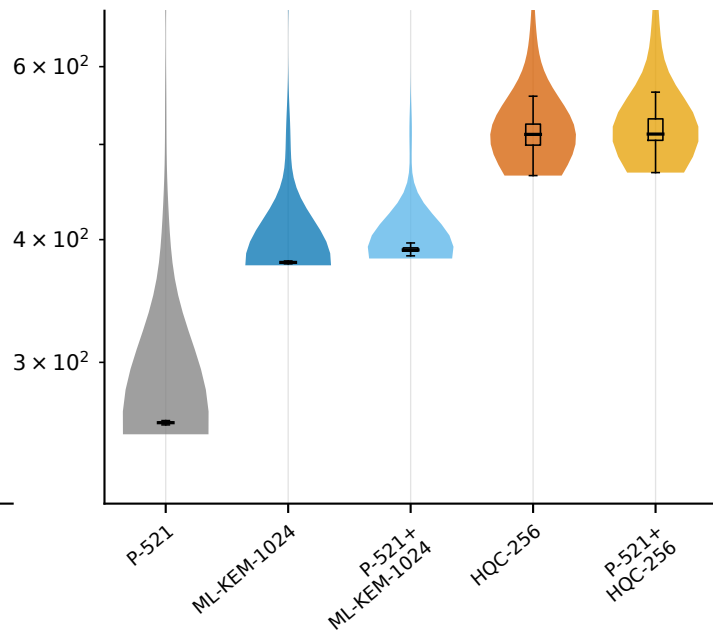


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	279.1	39.9	1.33k	309.3	59.0	1.63k
ML-KEM-1024	279.6	45.8	1.32k	411.0	32.8	1.44k
P-521+ML-KEM-1024	303.3	47.4	1.33k	408.9	24.9	1.44k
HQC-256	532.0	24.8	1.55k	543.7	28.0	1.63k
P-521+HQC-256	534.7	24.3	1.58k	550.2	27.6	2.23k